Local Spacing-Aware Hungarian Matching for Stable Point-Supervised Crowd Counting

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Abstract. Point-supervised crowd counting and localization enable end-to-end training by casting prediction as set matching with one-to-one Hungarian assignment. However, point annotations provide no explicit scale cues, making distance-based assignment prone to neighbor-induced ambiguity in congested regions, where nearby ground-truth points compete for similar proposals and corrupt supervision. We propose Local Spacing-Aware Hungarian Matching (SAH-matcher), a drop-in replacement that derives a local spacing prior from k-nearest-neighbor distances and performs per-target rescaling of the geometric cost, tightening the feasible matching range in dense areas while remaining tolerant in sparse ones. To quantify assignment behavior, we introduce Competitive Ambiguity Score (CAS) and Hijacking Rate (HR) for within-epoch ambiguity and severe hijacking failures, and combine them with Instability Rate (IR) to measure cross-epoch consistency. Experiments on multiple benchmarks show improved assignment reliability and training stability, leading to consistent gains in counting and localization with modest overhead, and strong plug-and-play generalization across point-based frameworks. Code is available at `https://github.com/kaijiang77/SAHCC`.

Keywords: Crowd Counting · Point Supervision · Hungarian Matching · Assignment Stability

1 Introduction

Crowd counting and localization are fundamental problems in computer vision, with applications in public safety, smart-city analytics, and transportation management [1, 2, 15, 19, 28, 30, 38, 42, 45, 47]. Real-world scenes remain challenging because strong perspective-induced scale variation and highly non-uniform density [11, 46], together with severe occlusion, background clutter, and imaging degradations [34, 40], often make nearby instances difficult to separate, thereby increasing ambiguity in point-level localization and matching.

Existing approaches are broadly categorized into map-based and localization-based paradigms. Map-based methods regress dense maps and integrate them

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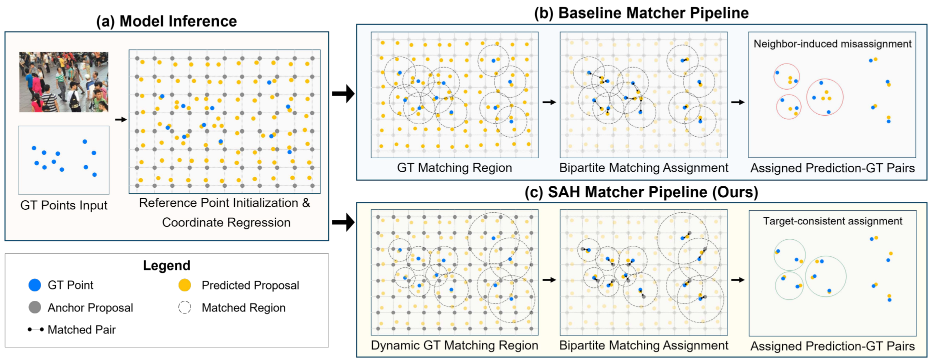

Fig. 1: Comparison of matching pipelines in point-based counting. Baseline Hungarian matching uses a fixed geometric scale, inducing a fixed GT matching region and ambiguous local competition in dense areas. SAH-matcher modulates the geometric cost by local spacing, yielding a target-specific matching region and more consistent one-to-one assignments.

for counting, but localization often relies on heuristic post-processing [14, 46]. Localization-based methods predict instance-level outputs more explicitly; under point supervision, point-based set prediction with one-to-one Hungarian matching has become a natural choice for end-to-end localization [5, 35].

However, point annotations provide only centers and therefore contain no explicit scale or geometric cues. Consequently, Hungarian assignment in existing point-based frameworks typically combines classification confidence with a fixed-form, scale-agnostic distance cost. In congested regions, neighboring ground-truth points can yield near-tie geometric costs to the same proposals, increasing the chance of *neighbor-induced misassignment* and corrupting set-based supervision. This failure mode is illustrated in Fig. 1.

Motivated by this observation, we propose **Local Spacing-Aware Hungarian Matching** (SAH-matcher), a drop-in replacement for standard Hungarian assignment in point-based frameworks. Our key insight is that, although point annotations do not provide explicit scales, they still imply a reliable local spacing prior: the local density and effective scale around each ground-truth point can be approximated by its k-nearest-neighbor distances. SAH-matcher leverages this prior to adaptively rescale the geometric cost for each target, thereby reducing dense-region ambiguity while preserving global one-to-one matching. As also shown in Fig. 1, this spacing-aware modulation yields a target-specific matching range, making the assignment process more selective in congested regions without changing the global Hungarian formulation.

Beyond improving final MAE and RMSE, we further argue that assignment behavior itself should be explicitly diagnosed rather than only inferred from final metrics. To this end, we introduce process-level diagnostics for assignment dynamics. We use Competitive Ambiguity Score (CAS) and Hijacking Rate (HR) to characterize within-epoch ambiguity and *neighbor hijacking* failure cases, where a matched proposal is closer to a neighboring target than to its assigned one, and combine them with Instability Rate (IR) [5] to capture cross-epoch assignment consistency. These diagnostics provide a more interpretable view of how assignment quality and stability evolve during training.

Our contributions are summarized as follows:

- We propose **SAH-matcher**, which injects a lightweight local spacing prior into Hungarian matching through target-wise geometric cost rescaling, effectively reducing dense-region competition without changing the global one-to-one formulation.
- We introduce **Competitive Ambiguity Score (CAS)** and **Hijacking Rate (HR)**, and combine them with **Instability Rate (IR)** to provide a process-level diagnostic view of assignment ambiguity, severe failure cases, and stability in point-based crowd counting.
- Extensive experiments on standard benchmarks demonstrate consistent gains in counting and localization, as well as strong plug-and-play generalization across different point-based frameworks.

2 Related Work

We review representative paradigms for crowd counting and localization, and then focus on prior efforts that improve assignment quality in point-based one-to-one frameworks.

2.1 Map-based Approaches

Map-based methods typically convert point annotations into pseudo supervisory signals, such as density maps or response maps, and train a regressor whose integral yields the crowd count [11, 14, 46]. A large body of work improves this paradigm through stronger architectures and better learning objectives [9,17,20, 24,27,33,39,43]. These methods are often effective for counting, but localization is usually obtained through heuristic post-processing, for example peak extraction or thresholding [1,12]. As a result, the optimization target is not naturally aligned with end-to-end point localization, especially in highly congested regions where overlapping responses reduce separability.

2.2 Localization-based Approaches

Localization-based methods aim to predict instance-level outputs more explicitly. Detection-based methods cast crowd counting as object detection, regress boxes or center-scale representations, and suppress duplicates through non-maximum suppression [29]. When only point annotations are available, pseudo box scales are often estimated from local geometry and refined during training [25, 32]. Although detection-based formulations provide explicit localization, heavy overlap in dense crowds makes duplicate suppression brittle and increases the difficulty of scale estimation [8, 41].

Point-based methods provide a more natural alternative under point supervision. They predict a set of point proposals and establish one-to-one correspondences to ground-truth points through Hungarian matching, enabling end-to-end training and direct point localization [5, 18, 23, 35]. This formulation removes the need for bounding boxes and non-maximum suppression, but it also makes the final performance highly dependent on assignment correctness and stability.

2.3 Point-based One-to-One Counting and Assignment Stabilization

P2PNet [35] introduces DETR-style set prediction [4] into point-based crowd counting by matching proposals to annotated centers through one-to-one Hungarian assignment. This formulation is simple and elegant, but point annotations provide only center locations and do not contain explicit scale cues. Consequently, the geometric term in Hungarian matching is usually implemented as a globally weighted point-to-point distance, which is inherently scale-agnostic and becomes more ambiguous in congested regions.

Several subsequent works improve this assignment process from different perspectives. CLTR [18] enhances matching by incorporating neighborhood context into the cost. APGCC [5] further highlights the instability of proposal-to-target matching, introduces auxiliary point guidance, and proposes Instability Rate for measuring cross-epoch assignment consistency. P2R [21] weakens direct competition among neighboring targets by replacing point-to-point assignment with a point-to-region formulation, where supervision is aggregated within local regions rather than enforced by strict global one-to-one matching. These efforts improve matching stability to different extents, but dense-region competition under a global geometric cost remains a key bottleneck.

Our work follows this line and focuses directly on the matching cost itself. Instead of changing the global assignment formulation, we inject a lightweight local spacing prior into Hungarian matching so that the geometric cost can adapt to local density around each target. This design preserves global one-to-one optimality while reducing ambiguity in congested regions, which is the central issue addressed by SAH-matcher.

3 Preliminaries: Point-based One-to-One Counting

We briefly review the standard set-prediction pipeline used in point-based crowd counting, focusing on the assignment and supervision components most relevant to our method [4, 35].

3.1 Problem Setup

Given an image $I \in \mathbb{R}^{H\times W\times 3}$ with point annotations

$$\mathcal{G} = \{\mathbf{g}_i\}_{i=1}^{M}, \quad \mathbf{g}_i \in \mathbb{R}^2, \tag{1}$$

a point-based model predicts a set of Q proposals

$$\hat{\mathcal{Q}} = \{(\hat{c}_j, \hat{\mathbf{p}}_j)\}_{j=1}^{Q}, \tag{2}$$

where $\hat{c}_j \in [0,1]$ denotes the foreground confidence and $\hat{\mathbf{p}}_j \in \mathbb{R}^2$ denotes the predicted point coordinate. Typically, $Q \geq M$, so not all proposals are expected to match a ground-truth point.

3.2 Hungarian Assignment

Training establishes one-to-one correspondences between proposals and ground-truth points through Hungarian matching. Let π be an injective mapping from GT indices to proposal indices. The optimal assignment is obtained by minimizing the total matching cost

$$\hat{\pi} = \arg\min_{\pi} \sum_{i=1}^{M} C_{\pi(i),i}, \tag{3}$$

where $C_{j,i}$ measures the cost of assigning proposal j to target i.

A common formulation combines classification confidence with geometric proximity:

$$C_{j,i} = \lambda_{\mathrm{cls}} C_{j,i}^{\mathrm{cls}} + \lambda_{\mathrm{geo}} C_{j,i}^{\mathrm{geo}}, \tag{4}$$

with

$$C_{j,i}^{\mathrm{cls}} = -\hat{c}_j \quad (\text{or } -\log \hat{c}_j), \qquad C_{j,i}^{\mathrm{geo}} = \|\mathbf{u}_j - \mathbf{g}_i\|_2 \,. \tag{5}$$

Here $\mathbf{u}_j$ denotes the reference point used for matching. In standard point-based formulations, Hungarian matching is usually defined on the regressed proposal coordinates $\hat{\mathbf{p}}_j$ [5, 35].

3.3 Set-based Supervision

Once the assignment $\hat{\pi}$ is obtained, matched proposals are treated as foreground and the remaining proposals are treated as background:

$$\mathcal{Q}^{+} = \{\hat{\pi}(i) \mid i = 1, \ldots, M\}, \qquad \mathcal{Q}^{-} = \{1, \ldots, Q\} \setminus \mathcal{Q}^{+}. \tag{6}$$

Training then applies binary classification supervision to both matched and unmatched proposals, together with point regression on matched pairs. A typical formulation is

$$\mathcal{L}_{\mathrm{cls}} = \frac{1}{Q}\Big(\sum_{j \in \mathcal{Q}^{+}} \mathrm{BCE}(\hat{c}_j, 1) + \lambda_{\mathrm{neg}} \sum_{j \in \mathcal{Q}^{-}} \mathrm{BCE}(\hat{c}_j, 0) \Big), \tag{7}$$

and

$$\mathcal{L}_{\mathrm{loc}} = \frac{1}{M} \sum_{i=1}^{M} \left\| \hat{\mathbf{p}}_{\hat{\pi}(i)} - \mathbf{g}_i \right\|_2^2 . \tag{8}$$

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{loc}}\mathcal{L}_{\text{loc}}. \tag{9}$$

This formulation makes the quality of Hungarian assignment central to training: the assignment directly determines which proposals receive positive supervision and which point targets they are optimized toward.

4 Method

4.1 Overview

Under point supervision, Hungarian assignment typically relies on a fixed-form geometric cost shared by all targets, which becomes ambiguous in crowded scenes with large local scale variation. In dense areas, neighboring ground-truth points often induce near-tie costs to the same proposals, increasing neighbor-induced misassignment and destabilizing set-based supervision; similar instability has been observed in DETR-style training [16].

To address this issue, we propose SAH-matcher, which augments Hungarian assignment with a local spacing prior for each ground-truth point. Figure 1 overviews the two pipelines: compared with the baseline matcher, SAH-matcher yields a dynamic, target-specific GT matching region by modulating the geometric cost with local spacing priors. Importantly, SAH-matcher preserves the global one-to-one assignment formulation and only modifies how the geometric cost is computed.

4.2 Local Spacing Prior via k-Nearest Neighbors

We use local point spacing as a lightweight proxy for local density and effective object scale [3,25,46]. Given the ground-truth set $\mathcal{G} = \{\mathbf{g}_i\}_{i=1}^{M}$, let $\mathcal{N}_k(i)$ denote the k nearest neighbors of $\mathbf{g}_i$ (excluding itself). We define the local spacing of target i as

$$d_i = \frac{1}{k} \sum_{\mathbf{g} \in \mathcal{N}_k(i)} \|\mathbf{g} - \mathbf{g}_i\|_2 \,. \tag{10}$$

Smaller d_i indicates a more crowded neighborhood (stronger competition), while larger d_i corresponds to relatively sparse regions. We transform d_i consistently under geometric augmentations so it stays in the same coordinate system as the matching reference.

4.3 SAH-matcher: Spatially Adaptive Hungarian Matching

Reference for matching. In standard point-based formulations, Hungarian matching is typically defined on regressed proposal coordinates. We unify both settings by denoting the matching reference as $\mathbf{u}_j$: $\mathbf{u}_j = \hat{\mathbf{p}}_j$ for pred-ref matching, and $\mathbf{u}_j = \mathbf{a}_j$ for anchor-ref matching. In our SAH-based implementation, we compute geometric matching costs on the fixed anchor reference ($\mathbf{u}_j = \mathbf{a}_j$), while the localization loss is still applied to the regressed coordinate $\hat{\mathbf{p}}_j$.

Algorithm 1 SAH-matcher

Require: Proposals $\{(\hat{c}_j, \mathbf{u}_j)\}_{j=1}^{Q}$, GT points $\{\mathbf{g}_i\}_{i=1}^{M}$, spacing prior $\{d_i\}_{i=1}^{M}$
Ensure: Matched indices $(I_{\text{pred}}, I_{\text{gt}})$
for $i = 1$ to M **do**
  $\alpha_i \leftarrow \max(\alpha_{\min}, \kappa/(d_i + \epsilon))$
for $j = 1$ to Q **do**
  for $i = 1$ to M **do**
    $C^{\text{cls}}_{j,i} \leftarrow -\hat{c}_j$
    $C^{\text{geo}}_{j,i} \leftarrow \alpha_i \cdot \|\mathbf{u}_j - \mathbf{g}_i\|_2$
    $C_{j,i} \leftarrow \lambda_{\text{cls}} C^{\text{cls}}_{j,i} + C^{\text{geo}}_{j,i}$
$(I_{\text{pred}}, I_{\text{gt}}) \leftarrow \text{Hungarian}(C)$
return $(I_{\text{pred}}, I_{\text{gt}})$

Spacing-aware geometric cost. We define the raw geometric error as $e_{j,i} = \|\mathbf{u}_j - \mathbf{g}_i\|_2$. In standard Hungarian matching, this term is globally weighted and therefore imposes the same effective matching scale for all targets. SAH instead introduces a target-specific modulation weight

$$\alpha_i = \max\left(\alpha_{\min}, \frac{\kappa}{d_i + \epsilon}\right), \tag{11}$$

where κ is a global scale factor and $\epsilon > 0$ avoids numerical instability when d_i approaches zero. We impose only a lower bound $\alpha_{\min}$ so that the geometric term does not become excessively weak in sparse regions, where large spacing values may otherwise enlarge the feasible matching range and destabilize assignment.

The spacing-aware geometric cost is then defined as

$$C^{\text{geo}}_{j,i} = \alpha_i \, \|\mathbf{u}_j - \mathbf{g}_i\|_2 \,. \tag{12}$$

Smaller spacing yields larger α_i and stronger geometric selectivity, while larger spacing preserves tolerance in sparse regions.

The final SAH-matcher cost keeps the same additive form, with the spacing-aware modulation already absorbed into the geometric term:

$$C_{j,i} = \lambda_{\text{cls}} C^{\text{cls}}_{j,i} + C^{\text{geo}}_{j,i} \,, \tag{13}$$

where the geometric scaling factor is absorbed into κ for simplicity.

4.4 Architecture Overview

As illustrated in Fig. 2, SAH-matcher replaces the baseline Hungarian matcher during training for proposal–GT assignment. The backbone, proposal generation, point head, and set-based losses remain unchanged, making SAH a drop-in assignment module.

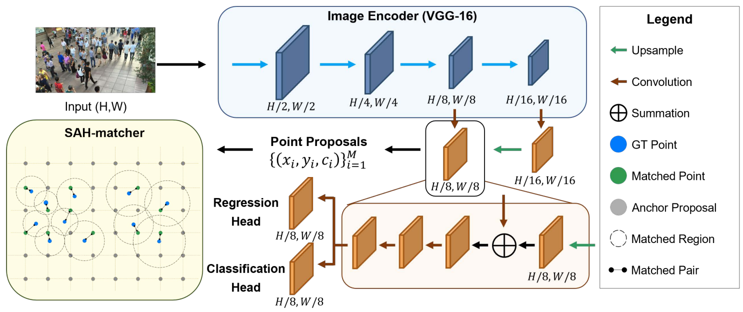

Fig. 2: Architecture overview of the training pipeline. SAH-matcher replaces the baseline Hungarian matcher for proposal–GT assignment, while keeping the backbone, proposal generation, and set-based supervision unchanged.

4.5 Assignment Quality Diagnostics

In point-based set-prediction frameworks, proposal–target assignment directly determines the supervision received by each proposal during training. Despite its central role, assignment behavior is usually evaluated only indirectly through final counting or localization performance. We therefore introduce process-level diagnostics that explicitly quantify within-epoch ambiguity and its severe failure cases.

Competitive Ambiguity Score (CAS). Given Hungarian assignment $\hat{\pi}$, define the matched distance and the nearest-competitor distance as

$$d_i^+ = \left\|\mathbf{u}_{\hat{\pi}(i)} - \mathbf{g}_i\right\|_2, \qquad d_i^- = \min_{k\neq i} \left\|\mathbf{u}_{\hat{\pi}(i)} - \mathbf{g}_k\right\|_2, \tag{14}$$

and compute

$$\mathrm{CAS}_i = \frac{2\,d_i^+}{d_i^+ + d_i^- + \epsilon}, \qquad \mathrm{CAS} = \frac{1}{M}\sum_{i=1}^{M} \mathrm{CAS}_i, \tag{15}$$

where $\epsilon > 0$ is a small constant. CAS approaches 0 when the matched proposal is well separated from competing targets, is around 1 under near-tie competition, and exceeds 1 when the proposal is closer to a competing target than to its assigned target.

Hijacking Rate (HR). To quantify severe failures, we summarize the case $\mathrm{CAS}_i > 1$ as

$$\mathrm{HR} = \frac{1}{M}\sum_{i=1}^{M} \mathbb{I}\big[d_i^+ > d_i^-\big] = \frac{1}{M}\sum_{i=1}^{M} \mathbb{I}\big[\mathrm{CAS}_i > 1\big], \tag{16}$$

which measures the fraction of targets that suffer from severe neighbor hijacking in an epoch. Here, a neighbor hijacking event refers to the case where the proposal matched to target i is actually closer to a competing neighboring target than to $\mathbf{g}_i$ itself, *i.e.*, $d_i^+ > d_i^-$. We regard this as a severe failure because the matched proposal is no longer even geometrically closest to its assigned target, indicating a cross-target takeover under local competition.

CAS and HR are computed from the same Hungarian matches used for optimization and with the same matching reference $\mathbf{u}_j$, so they reflect assignment behavior itself rather than downstream regression noise. Together, CAS captures continuous within-epoch ambiguity, while HR focuses on severe assignment failures.

5 Experiments

5.1 Datasets

We evaluate on SHHA, SHHB [46], UCF-QNRF [11], JHU-Crowd++ [34], and NWPU-Crowd [40], spanning dense-to-sparse regimes, strong scale variation, and unconstrained degradations.

5.2 Evaluation Metrics

Counting. We report mean absolute error (MAE) and root mean squared error (RMSE):

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{n}_i - n_i\right|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{n}_i - n_i\right)^2}. \tag{17}$$

Localization. We report precision (P), recall (R), and F_1 based on one-to-one matching between predicted points and ground-truth points [40]:

$$P = \frac{TP}{TP+FP}, \quad R = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R}. \tag{18}$$

A prediction is counted as a true positive if it is matched to a ground-truth point within a threshold σ. We use the standard fixed thresholds $\sigma \in \{4, 8\}$ on SHHA [5] and the adaptive threshold on NWPU ($\sigma = \sqrt{w^2 + h^2}/2$) [40].

5.3 Training Details

We train with AdamW [26] using grouped learning rates (10^{-4} non-backbone, 10^{-5} backbone) and batch size 8. Following P2PNet [35], we use stride $s = 8$ with $K = 4$ references on ShanghaiTech and $s = 4$ with $K = 1$ otherwise; matching costs are computed on anchors while regression losses are applied to predicted points. Unless stated otherwise, we set $\lambda_{\text{cls}} = 1$, $\lambda_{\text{loc}} = 2 \times 10^{-4}$, and SAH uses $\kappa = 1.2$, $\alpha_{\min} = 0.03$.

Table 1: Main results on crowd counting (MAE/RMSE). Best is in bold and second best is underlined for each metric.

Method	SHHA		SHHB		UCF-QNRF		JHU-Crowd++		NWPU(T)	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Detection-based										
LSC-CNN [32]	66.40	117.00	8.10	12.70	120.50	218.20	–	–	–	–
NoisyCC [36]	61.90	99.60	7.40	11.30	85.80	150.60	67.70	258.50	96.90	534.20
TopoCount [1]	61.20	104.60	7.80	13.70	89.00	159.00	60.90	267.40	107.80	438.50
Map-based										
CSRNet [17]	68.20	115.00	10.60	16.00	–	–	85.90	309.20	121.30	387.80
CAN [24]	62.30	100.00	7.80	12.20	107.00	183.00	100.10	314.00	106.30	386.50
S-DCNet [43]	58.30	95.00	6.70	10.70	104.40	176.10	–	–	90.20	370.50
DM-Count [39]	59.70	95.70	7.40	11.80	85.60	148.30	–	–	88.40	388.60
UOT [27]	58.10	95.90	6.50	10.20	83.30	142.30	60.50	252.70	87.80	387.50
ChfL [33]	57.50	94.30	6.90	11.00	80.30	137.60	57.00	235.70	76.80	343.00
MAN [20]	56.80	90.30	–	–	77.30	**131.50**	**53.40**	**209.90**	76.50	323.00
HMoDE+REL [9]	54.40	87.40	6.20	9.80	–	–	55.70	214.60	73.40	331.80
Point-based										
P2PNet [35]	52.74	85.06	6.25	9.90	85.32	154.50	–	–	83.28	553.92
CLTR [18]	56.90	95.20	6.50	10.60	85.80	141.30	59.50	240.60	74.30	333.80
PET [23]	49.34	78.77	6.19	9.69	79.53	144.32	58.50	238.00	74.40	328.50
APGCC [5]	48.80	76.70	**5.60**	**8.70**	80.10	136.60	54.30	225.90	71.40	**284.40**
P2R [21]	51.02	79.68	6.17	9.84	83.26	138.11	58.83	253.10	–	–
Ours	**47.23**	**75.17**	6.14	9.50	**76.91**	135.92	58.32	250.62	**68.90**	306.70

Data preprocessing and augmentation. We apply random scaling [0.7, 1.3], random cropping (128×128 for ShanghaiTech; 256×256 otherwise), horizontal flipping, and photometric augmentations [6]; for UCF-QNRF/JHU/NWPU we cap the longer side at 2048 (aspect ratio preserved).

5.4 Main Results

Crowd counting. Table 1 reports MAE/RMSE of representative detection-based [1,32,36], map-based [9,17,20,24,27,33,39,43], and point-based [5,18,21,23,35] methods. SAH-matcher achieves consistently strong results and is particularly competitive on dense or strongly scale-varying benchmarks (e.g., SHHA, UCF-QNRF, and NWPU), where near-tie geometric costs amplify local competition in Hungarian assignment.

This pattern is consistent with our motivation: SAH mainly helps when dense-region near-tie competitions make assignments fragile. When scenes are sparser, such competition is weaker and the gain shrinks (SHHB); when distribution shift and degradations dominate the error budget, refining assignment alone offers limited headroom (JHU-Crowd++).

Crowd localization. We further evaluate localization on SHHA and NWPU using Precision, Recall, and F_1 (Tables 2 and 3). SAH-matcher yields consistently strong localization on both benchmarks, with larger gains under stricter thresholds, which are more sensitive to cross-target misassignment and dense-region ambiguity [40].

Table 2: Evaluation of crowd localization on SHHA [46] dataset.

Method	$\sigma = 4$			$\sigma = 8$		
	F(%)	P(%)	R(%)	F(%)	P(%)	R(%)
Map-based						
LOBB [13]	25.9	34.9	20.7	53.9	67.6	44.8
Detection-based						
LCFCN [31]	32.5	43.3	26.0	56.3	75.1	45.1
LSC-CNN [32]	32.6	33.4	31.9	62.4	63.9	61.0
TopoCount [1]	41.1	41.7	40.6	73.6	74.6	72.7
Point-based						
P2PNet [35]	40.6	41.5	39.8	74.6	76.2	73.1
CLTR [18]	43.2	43.6	42.7	74.2	74.9	73.5
APGCC [5]	48.7	49.2	48.3	**78.4**	**79.1**	**77.7**
Ours	**48.9**	**49.3**	**48.6**	78.1	78.6	77.6

Table 3: Evaluation of crowd localization on NWPU [40] dataset.

Method	σ_l (large)			σ_s (small)		
	F(%)	P(%)	R(%)	F(%)	P(%)	R(%)
Detection-based						
TinyFaces [10]	56.7	52.9	61.1	52.6	49.1	56.6
TopoCount [1]	63.7	65.1	62.4	–	–	–
Map-based						
RAZ [22]	59.8	66.6	54.3	57.6	47.0	51.7
GL [37]	66.0	**80.0**	56.2	58.7	71.1	50.0
D2CNet [7]	70.0	74.1	66.2	63.2	67.0	59.8
AutoScale [44]	67.3	57.4	62.0	–	–	–
Point-based						
P2PNet [35]	71.2	72.9	69.5	67.5	68.4	66.6
CLTR [18]	68.5	69.4	67.6	59.1	59.9	58.3
PET [23]	74.2	75.2	73.2	67.5	68.4	66.6
APGCC [5]	76.4	79.2	73.6	68.9	71.5	66.5
Ours	**77.7**	77.8	**77.6**	**72.1**	**72.1**	**72.0**

On SHHA, SAH-matcher achieves the best F_1 under $\sigma = 4$ and remains comparable under $\sigma = 8$. On NWPU, improvements are more pronounced under the stricter setting (σ_s), showing simultaneous gains in Precision and Recall, indicating more reliable proposal-to-target alignment across a wide density range.

5.5 Generalization as a Plug-in Matcher

To evaluate the portability of SAH-matcher, we replace the original matcher in several representative point-based frameworks while keeping the remaining training pipeline unchanged. Results are reported in Table 4. This comparison is intended to assess cross-framework transferability under a controlled setting, rather than the best achievable performance of each method after full retuning; for each method's best-reported results, please refer to Table 1. Overall, SAH-matcher improves MAE in most settings, with larger gains on denser benchmarks where near-tie geometry induces stronger local competition.

Integration details. For CLTR [18], we retain its KMO-based Hungarian matcher and context-based auxiliary cost, and apply SAH only to the distance-related term. For P2R [21], we evaluate only its fully supervised branch to avoid confounding effects from pseudo-labeling and teacher–student components.

5.6 Assignment Dynamics Diagnostics

CAS/HR by local spacing and IR over epochs. Fig. 3 shows that SAH mainly benefits small-d_i bins (congested regions): CAS drops from 0.63 to 0.44 and HR from 0.16 to 0.02, indicating reduced near-tie ambiguity and far fewer severe hijacking events. IR is also consistently lower throughout training and decreases from 0.123 to 0.059, reflecting more stable proposal–target assignments across epochs.

Table 4: Portability of SAH-matcher across point-based frameworks (MAE/RMSE). Avg. Δ is computed as (+SAH − Baseline) over available benchmarks; negative values indicate improvements. Within each method, the better result between Baseline and +SAH is highlighted in bold (lower is better).

Method	Variant	SHHA		SHHB		QNRF		JHU		Avg. Δ	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	ΔMAE	ΔRMSE
P2PNet [35]	Baseline	54.70	89.06	6.29	10.18	93.81	164.21	62.23	269.70	–	–
	+SAH	**50.30**	**81.86**	**6.11**	**9.97**	**85.81**	**151.43**	**60.85**	**264.30**	-3.49	-6.40
CLTR [18]	Baseline	66.53	114.74	7.53	13.16	93.49	166.72	64.17	**257.45**	–	–
	+SAH	**65.20**	**108.89**	**7.49**	**12.62**	**91.11**	**154.90**	**62.74**	258.86	-1.29	-4.20
PET [23]	Baseline	51.90	83.64	**6.56**	**10.23**	95.66	**162.57**	59.98	252.52	–	–
	+SAH	**49.76**	**80.99**	6.76	10.95	**88.99**	164.32	**59.05**	**250.93**	-2.39	-0.44
P2R [21]	Baseline	51.02	79.68	6.95	11.54	93.50	168.48	**62.23**	272.66	–	–
	+SAH	**50.67**	**78.50**	**6.68**	**10.51**	**88.75**	**154.74**	63.45	**267.91**	-1.03	-5.18

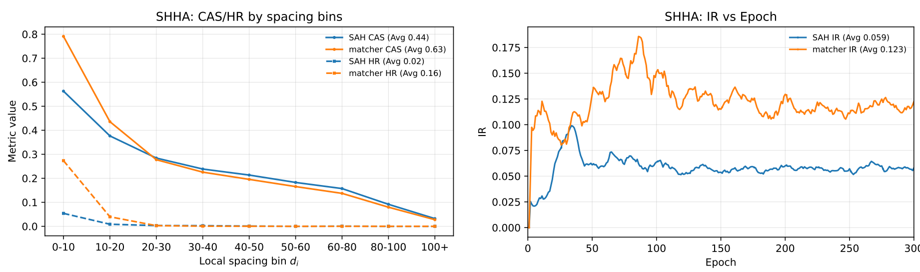

Fig. 3: Assignment diagnostics on SHHA. Left: CAS/HR over spacing bins (avg over last 10 epochs), with CAS 0.63→0.44 and HR 0.16→0.02. Right: IR over epochs (moving average), with avg IR 0.123→0.059.

Overall, these diagnostics show that SAH improves assignment quality precisely where standard Hungarian matching is most vulnerable—densely packed regions with strong local competition—while also making proposal–target assignment more consistent over the course of training. To further illustrate these effects, Fig. 4 provides qualitative comparisons of two representative assignment failures—neighbor hijacking and neighbor-induced misassignment—where the baseline matcher exhibits cross-target takeover and ambiguous pairing in congested regions, while SAH-matcher restores target-consistent one-to-one assignments.

5.7 Ablation Study

Effectiveness of SAH-matcher. We isolate SAH by replacing only the matcher and keeping the backbone, proposal generation, point head, and losses unchanged. Table 5 shows that SAH's per-target rescaling consistently outperforms a global sweep of the geometric-cost weight, indicating that the gain comes from spacing-aware modulation rather than tuning a single coefficient.

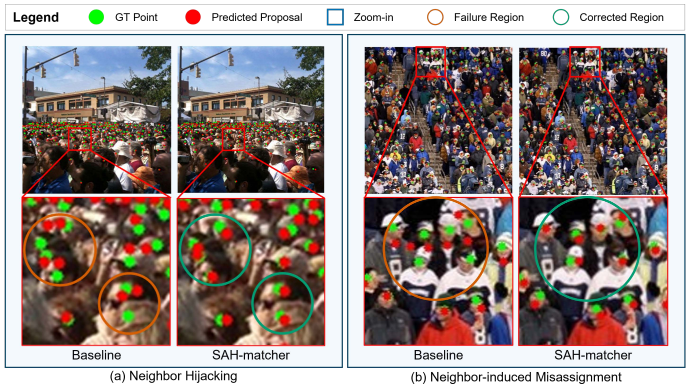

Fig. 4: Qualitative assignment failures and SAH-matcher remedies. (a) Neighbor hijacking: baseline matching exhibits cross-target takeover in congested regions, while SAH restores target-consistent one-to-one assignments. (b) Neighbor-induced misassignment: baseline matching yields ambiguous cross-target pairing, whereas SAH suppresses such misassignments and recovers consistent matches.

Table 5: Global geometric-cost weight sweep ($\lambda_{\text{geo}} = w$) vs. SAH dynamic per-target rescaling on SHHA (MAE/RMSE).

Variant	MAE ↓	RMSE ↓
Fixed $w = 0.01$	51.82	85.64
Fixed $w = 0.04$	50.44	79.87
Fixed $w = 0.05$ (baseline)	50.06	81.57
Fixed $w = 0.07$	50.53	81.68
Dynamic per-target (SAH)	**47.23**	**75.17**

Table 6: Effect of matching reference $\mathbf{u}_j$ (anchor points vs. predicted points) on SHHA (MAE/RMSE).

Variant	MAE ↓	RMSE ↓
SAH + predicted points	51.63	86.84
SAH + anchor points	**47.23**	**75.17**
Baseline matcher + predicted points	50.89	87.27
Baseline matcher + anchor points	50.70	81.76

The gain is also not explained by switching the matching reference. In Table 6, changing the baseline reference from predicted points to anchors yields only a marginal MAE change (50.89→50.70), while SAH under the same anchor reference achieves 47.23 MAE and improves over the best fixed-w setting. We therefore treat anchors as a stabilizing reference space for cost evaluation; applying target-wise modulation on noisy regressed points can create early prediction–assignment feedback, consistent with prior analyses of bipartite matching instability [16].

Hyper-parameters. We ablate the neighborhood size k (Eq. (10)) and the lower bound $\alpha_{\min}$ (Eq. (11)). Table 7 shows robust performance across a reasonable range, with the best setting at $k = 4$ and $\alpha_{\min} = 0.03$.

Computation. SAH-matcher adds only a lightweight target-wise modulation to the geometric cost, while leaving the Hungarian solver itself unchanged. The spacing prior is precomputed offline, so the extra runtime comes mainly from computing α_i and rescaling the geometric-cost matrix. Table 8 reports training-

Table 7: Hyper-parameter sensitivity of SAH-matcher on SHHA (MAE/RMSE): neighborhood size k and lower bound $\alpha_{\min}$.

Effect of k in the spacing prior

k	MAE ↓	RMSE ↓
1	50.10	81.81
2	48.12	78.98
3	48.85	82.70
4	**47.23**	**75.17**
5	48.73	75.43
6	50.35	79.67
7	49.74	81.14

Effect of lower bound $\alpha_{\min}$

$\alpha_{\min}$	MAE ↓	RMSE ↓
0.01	51.82	85.64
0.02	50.79	82.40
0.03	**47.23**	**75.17**
0.04	48.42	78.81
0.05	48.12	78.98
0.06	49.28	78.86
0.07	48.64	79.38

Table 8: Training-time overhead of SAH-matcher on SHHA under identical settings. We report end-to-end iteration time, matcher-only time, and relative overhead.

Matcher	End-to-end (ms/iter) ↓	Matcher-only (ms/iter) ↓	Overhead (%) ↓
Baseline matcher	104.44	21.46	–
SAH-matcher	115.41	33.28	10.50

time overhead on SHHA under identical settings, measured with warm-up and synchronized CUDA timing. Overall, the added cost is modest at the end-to-end level, indicating that SAH improves assignment quality without materially changing the training recipe.

6 Conclusion

We study assignment ambiguity and instability in point-supervised crowd counting, where a globally weighted and scale-agnostic Hungarian cost can become unreliable in congested regions. To address this issue, we introduce SAH-matcher, a lightweight drop-in replacement that uses a kNN-based local spacing prior to adaptively rescale the geometric matching cost for each target, while preserving global one-to-one assignment.

Beyond final counting accuracy, we further emphasize that assignment behavior itself should be explicitly diagnosed. To this end, we introduce CAS and HR, and use them together with IR to characterize within-epoch ambiguity, severe failure cases, and cross-epoch assignment consistency.

Extensive experiments show that SAH-matcher improves both counting and localization, with the clearest gains on dense or strongly scale-varying benchmarks, and generalizes well as a plug-in module across different point-based frameworks. Overall, the results support the view that making Hungarian matching aware of local spacing is an effective and practical way to improve assignment reliability in point-based crowd counting.

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